# Unnamed_Unit

# Unit 3 Basics of Debugging

## PDF Documents

### PDF Document 1: 17318416834448-Chapter 1_unit 3 Basics of Debugging.pdf

This PDF document is included in the unit materials.

## YouTube Links

# UNIT 3 BASICS OF DEBUGGING


Welcome to the intriguing world of debugging! Imagine you're a detective, and your mission is to solve mysteries. But instead of solving crimes, you're solving puzzles hidden inside computer programs, called "bugs." Even without much programming experience, you can become a great bug detective by using logic and observation, skills you use every day!

## Introduction to Debugging

### WHAT ARE BUGS, REALLY?

Just like a small piece of a puzzle placed incorrectly can spoil the whole picture, a tiny mistake in a computer program can cause it to not work properly. Debugging is how we find and fix these mistakes, making sure our programs do exactly what we want them to.

### REAL-WORLD DEBUGGING: THE MARS CLIMATE ORBITER

Back in 1999, NASA sent a space robot called the Mars Climate Orbiter to study the weather on Mars. Everyone was excited to learn more about the Red Planet. But there was a tiny mistake in the program that controlled the orbiter.

(NASA launched the Mars Climate Orbiter today (Dec. 11, 1998) from Cape Canaveral, Fla)

inches, you used a different kind of ruler with centimeters, without realizing they're different. Because of this mix-up, the Mars Climate Orbiter didn't go where it was supposed to. Instead of safely entering orbit around Mars, it got too close to the planet and was lost. This mistake was a big deal because the mission cost a lot of money, and scientists lost a chance to learn more about Mars. This story teaches us how important it is to check our work and be very clear when we give instructions, especially when writing programs for computers or space robots. A small mistake, like mixing up inches and centimeters, can lead to big problems. That's why debugging and paying attention to details are super important skills, not just in programming but in many areas of life.

## Understanding Bugs with Daily Analogies

Bugs in computer programs are just like little hiccups in our daily routines that need fixing.

### THE CASE OF THE MISSING HOMEWORK:

Imagine you always put your homework in your backpack, but one day, you find it's not there. You retrace your steps, think about where it could be, and check those places. This is just like debugging! You're looking for the mistake (the "bug") that caused the problem (missing homework).

### THE RECIPE GONE WRONG:

Think about baking a cake but accidentally using salt instead of sugar. The cake doesn't taste good, right? Finding out why is like debugging. You check the recipe (the "program") to find where things went wrong.

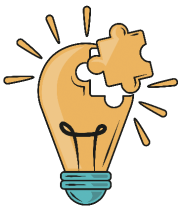
## Strategies for Debugging Algorithms

Now that we know what bugs are, how do we fix them? Here are some detective tools and strategies that can help.

### ASK QUESTIONS:

Just like detectives ask questions to solve mysteries, ask yourself questions about the problem in the program. When did it start? What changed?

### TRACE YOUR STEPS:



Pay attention to the details. In programming, small things like a misplaced letter or symbol can cause big problems.

### ASK QUESTIONS:

**The Mistake:** Rushing the Recipe
**First Attempt:** The operations were performed from left to right, like reading a book, resulting in: 5 * 8 - 8 = 40 - 8 = 32.
**The Error:** Just like adding the frosting before baking the cake, the mistake here was not following the correct order of operations.

### DEBUGGING CHART

| Step | Expression | Note |
|---|---|---|
| 1 | 3 + 2 * 8 - 8 | Original Problem |
| 2 | 3 + (2 * 8) - 8 | Highlight Multiplication |
| 3 | 3 + 16 - 8 | Perform Multiplication |
| 4 | 19 - 8 | Add |
| 5 | 11 | Subtract |

## Debugging in Action

Let's put our detective hats on and try debugging some algorithms with common everyday activities.

### ACTIVITY 1: THE CASE OF THE LOST HOMEWORK

An algorithm for organizing your backpack has a bug that leads to lost homework. Review the steps of the algorithm to find and fix the mistake. Where is my lost homework?

**EXAMPL**

**TRACING BACK THE STEPS!**

| | |
|---|---|
| at 16:00 on March 17th, 2024 | the teacher handed out the homework |
| at 20:15 on March 17th, 2024 | the homework was completed and shared with Hugh |
| at 11:00 on March 18th, 2024 | i reviewed the homework with hugh in the classroom |
| at 12:15 on March 18th, 2024 | Hugh took the homework, saying that he would put it in my bag again |
| at 16:00 on March 18th, 2024 | the homework was found lost |

### CONCLUSION: EMBRACE THE CHALLENGECONCLUSION: EMBRACE THE CHALLENGE

Remember, finding and fixing bugs in algorithms isn't just a crucial skill for programmers; it's a superpower for solving all kinds of problems. By becoming algorithm detectives, you're learning to think critically, creatively, and logically. Every bug you fix makes you a stronger problem solver!